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MetaFlux: A Mean-Field Phase–Amplitude Memory Substrate

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Disclaimer

This document is descriptive in nature and does not constitute a contractual commitment, license grant, or warranty. The material herein is provided for technical understanding and evaluation only.

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1. Executive Summary

MetaFlux is a deterministic, thermodynamically stable memory substrate based on globally coupled phase–amplitude oscillator dynamics. Unlike trained neural networks, MetaFlux does not rely on gradient descent, learned parameters, or backpropagation. Instead, it exploits mean-field dynamics to produce stable, macroscopic memory behavior from stochastic microscopic states.

MetaFlux is designed as an infrastructural component for intelligent systems, including large language model (LLM) agents, continual learning pipelines, and control architectures requiring stable, scalable memory without training instability.

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2. Motivation and Problem Statement

Modern AI systems rely heavily on trained neural architectures that suffer from several structural limitations:

- Training instability and sensitivity to hyperparameters
- Poor scalability of long-term memory
- Catastrophic forgetting in continual learning
- Opaque internal state dynamics
- High computational and energy cost during training

MetaFlux addresses these limitations by reframing memory as a dynamical physical process rather than a learned mapping.

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3. Conceptual Architecture

At its core, MetaFlux consists of a large population of coupled oscillatory units, each characterized by:

- A phase variable (phi)
- An amplitude variable (r)

The system is globally coupled through a mean-field feedback loop. Individual oscillators collectively determine a global order parameter, which in turn feeds back into the dynamics of each oscillator.

A high-level block diagram of the MetaFlux architecture is provided separately in `BLOCK_DIAGRAM.md`.

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4. Mean-Field Dynamics and Stability

The MetaFlux dynamics operate in a controlled stability regime defined by:

- Linear amplitude relaxation toward a target radius
- Phase coupling mediated by a global order parameter
- Explicit normalization by system size (K)

Empirical runs confirm high macroscopic coherence (R approximately 0.995) and stable trajectories across system sizes ranging from K = 10^5 to K = 10^7, with no divergence or numerical instability observed.

Importantly, MetaFlux dynamics do not depend on training history or learned weights.

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5. Scaling Behavior and Thermodynamic Limit

MetaFlux exhibits:

- O(K) time complexity
- O(K) memory complexity

- Stable behavior in the thermodynamic limit (K -> infinity)

As system size increases, microscopic fluctuations average out, and the system converges toward a deterministic mean-field trajectory.

This contrasts with attention-based memory mechanisms, which scale superlinearly and degrade under long-horizon workloads.

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6. Determinism, Noise, and Memory Semantics

Individual oscillators evolve with small random perturbations in phase and amplitude. These perturbations serve to break perfect symmetry and model realistic microscopic noise.

Given a fixed random seed and system parameters, the macroscopic trajectory is fully reproducible. Mean-field averaging ensures that stochastic microstates do not propagate instability to the global memory state.

MetaFlux thus functions as:

- A robust memory attractor
- A state persistence mechanism
- A controllable dynamical memory substrate

Noise is not eliminated; it is absorbed and averaged at the system level.

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7. Relationship to Neural Networks and LLMs

MetaFlux is not:

- A neural network
- A learning algorithm
- A replacement for transformers

Instead, MetaFlux complements LLMs by providing:

- Stable external memory
- Non-parametric state persistence
- Dynamical context storage

MetaFlux can be integrated alongside LLMs without modifying their architecture or training process.

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8. Intended Use Cases

Potential applications include:

- Long-horizon agent memory
- Continual learning stabilization
- Control systems requiring persistent internal state
- Research into non-trained cognitive substrates
- Hybrid symbolic–dynamical AI architectures

For example, an LLM agent could encode short-term context by briefly perturbing oscillator phases. The collective dynamics would then relax to a stable attractor representing persistent internal state without retraining or parameter updates.

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9. Non-Goals and Misinterpretations

MetaFlux is explicitly not intended to:

- Replace gradient-based learning
- Serve as a general-purpose optimizer
- Act as a black-box predictor
- Function as a drop-in neural model

It should be understood as infrastructure, not an end-user model.

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10. Current Status and Future Directions

MetaFlux v0.1.1 demonstrates:

- Stable mean-field dynamics
- Linear scalability to K approximately 10^7
- Deterministic macroscopic behavior under stochastic microstates

Future directions include:

- Controlled perturbation and recall mechanisms
- Multi-field or hierarchical coupling
- Formal information-theoretic characterization
- Hardware-aligned or neuromorphic implementations

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11. Conclusion

MetaFlux represents a shift in how memory can be constructed in intelligent systems: away from training-centric paradigms and toward stable dynamical substrates grounded in mean-field physics.

By decoupling memory stability from learning, MetaFlux opens new architectural space for scalable, interpretable, and robust AI systems.

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